# Refraction

Aim: To observe the relationship between angle of incidence and refraction

## Equipment:

- Ray box
- Single slit plate
- Lab pack (if needed)
- Protractor
- Pencil and ruler
- Paper

**Safety Information**

Filament bulb – risk of burns

Handle with extra care. Rinse any burns under cold running water for 10 minutes and inform the teacher immediately.

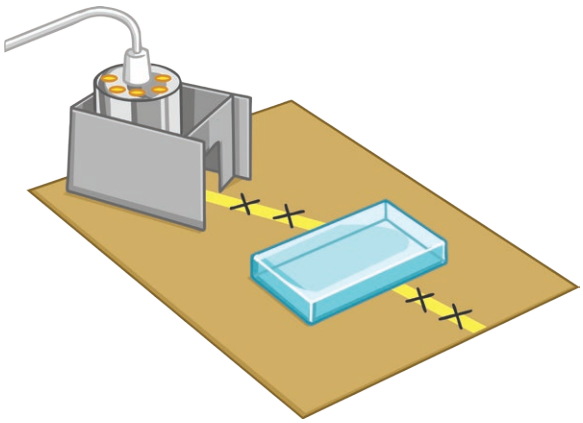


Figure 1. Set-up

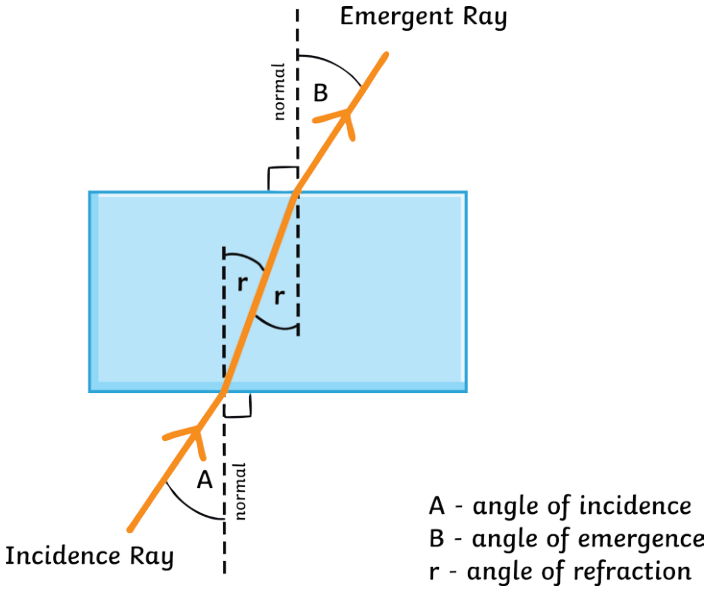



Figure 2. Measurements

## Method:

Step 1: Place the glass block flat in the middle of the paper and draw around it.

Step 2: Direct the light from the ray box onto one of the longer sides of the glass block, at a diagonal angle.

Step 3: Trace over the ray as it leaves the ray box and enters the glass box (the incidence ray). You can tidy up the line with the ruler afterwards.

Step 4: Trace over the ray as it leaves the glass block on the other side (the emergent ray). You can tidy up the line with the ruler afterwards.



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Step 5: Remove the glass block and join the incidence and emergent rays with a straight line.

Step 6: Use the ruler and protractor to mark normal lines at the point where the incidence ray enters the glass block, and the point where the emergent ray leaves the glass block.

Step 7: Measure the angle of incidence (A), angle of refraction (r) and angle of emergence (B). Record these in the table.

Step 8: Repeat steps 1 to 7 for at least four different angles of incidence.

Results:

| Angle of Incidence | Angle of Refraction | Angle of Emergence |
|---|---|---|
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Questions:

1. Describe the relationship between the angle of incidence and the angle of refraction.
2. Describe the relationship between the angle of incidence and the angle of emergence.
3. What is the name of the type of wave by which light is transmitted?
4. Explain refraction in terms of density and wave speed.



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# Answers:

1. The angle of incidence is greater than the angle of refraction; A > r.
2. The angles of incidence and emergence are equal.
3. Transverse wave (electromagnetic spectrum wave).
4. The velocity of the light wave is decreased as it enters a denser medium (the glass block). As it enters at an angle, the ray is 'bent' (refracted). When the light ray emerges into the original medium again (air) it resumes the same velocity and angle as before it entered the glass block.